Solar Forecasting Pipeline — Detailed Architecture

Team 2 · Sirmour Solar Plant, 5.1 MW · gemini-3.5-flash · amazon/chronos-bolt-small · pvlib Ineichen · ECMWF via Open-Meteo

Legend
Solid green = live in production · Dashed grey = built, currently OFF / unused in the forecast

1. Windy Satellite Capture

Playwright + Chromium (headless)

- Viewport 1280×720, zoom level 8
- Overlay: Satellite (VIS layer)
- 20-second animation clip recorded
- + solarpower & clouds still screenshots
- Fixed pixel-coordinate clicks for VIS/play buttons
- Uploaded to S3 (team bucket) after recording

↓ main pipeline · ⇢ built but unused (below)

Optical Flow (Farneback) — modules/vision/optical_flow.py

- Reads its own frames, converts to grayscale (required by Farneback — the only grayscale step in the project)
- Outputs cloud_cover_fraction, motion direction, speed_kmh
- Written as a JSON sidecar next to each clip

NOT fed into the forecast — measured to carry no forward-predictive signal (monsoon cloud here builds convectively, not by drifting in)

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2. Frame Extraction — OpenCV

- 10 frames, uniform sampling across the clip
- Saved in full colour (BGR) — NOT grayscale
- Skips the map-loading start & timeline loop-back

Frame Preprocessor (OpenCV) — modules/vision/frame_preprocessor.py

- Crop fixed Windy UI chrome
- CLAHE contrast enhancement on the L channel (LAB colour space) — keeps colour intact
- Optional fast denoising

OFF by default (preprocess=False) — built for the mentor's OpenCV question, not yet validated enough to ship

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3. Vision LLM

Provider: Google Gemini — model gemini-3.5-flash

- Free tier · config-swappable (make_vision_client)
- Alt provider: GPT-4o (OpenAI) / OpenRouter free vision models
- temperature 0.2 · max_output_tokens 8192
- Retry: 3 attempts, exponential backoff (5s, 10s) on HTTP 503
- Reads all 10 frames together, chronologically

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4. Weather Features (JSON)

- cloud_coverage_pct
- cloud_density (none / thin / moderate / thick)
- cloud_motion_direction + cloud_motion_speed
- moving_toward_plant (true/false)
- expected_change + minutes_until_change
- trend_next_2h / trend_2h_to_4h
- rain_probability_pct

- confidence (0.0 – 1.0)

Only trend_next_2h, trend_2h_to_4h, minutes_until_change and confidence actually reach the forecast — the rest is cached and unused

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merges with three additional inputs below

Previous Meter Generation

- Historical CSVs (*_SOLAR_INV.csv), 15-min interval
- Validate → outlier removal → ffill/bfill → time-align
- Pulled fresh from S3 before every run

Weather Forecast

- ECMWF IFS 0.25° via Open-Meteo (free, no key)
- Signal: forecast shortwave radiation (GHI) — NOT cloud-cover %
- Bias-corrected vs. last 5 finished days

Clear-Sky Physics

- pvlib — Ineichen clear-sky model
- Performance ratio: 0.80
- tilt = latitude · azimuth = 180° (assumed)

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5. Weighted Blend + Chronos

Numbers → Numbers — no LLM involved

- Chronos: amazon/chronos-bolt-small, CPU, zero-shot
- final_kt = 0.65 × weather_kt + 0.35 × base_kt
- base_kt = (0.20 × context_ratio) × chronos_kt + persistence
- Vision adjusts persistence only — capped ≈ ±4% of the final number
- final_kw = final_kt × clear-sky curve, clipped to 5100 kW

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6. Case-Based Correction

- k = 40 nearest analogue blocks (time-of-day, kt, horizon, forecast level)
- Applied at strength 0.5 — full strength (1.0) overfits and hurts

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7. Outputs

- <date>_<time>.csv — this run's 96-block forecast
- archive.csv — every run ever, append-only
- current_final_schedule.csv — actual (completed blocks) + forecast (blocks ahead)
- <date>_end_of_day_validation.json — after the 15:45 run
- Pushed to S3 → served by FastAPI dashboard + 3 JSON endpoints